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# Music Alarm Serial Communication Protocol

## Custom protocol (default)

(RS232/RS485)

### I. Basic communication parameters.

Baud rate: 9600    Data bits: 8    Stop bit: 1    Parity bit: None

### II. PLC master data (7 bytes in length):

| 0            | 1             | 2                   | 3       | 4        | 5                | 6        |
|--------------|---------------|---------------------|---------|----------|------------------|----------|
| Initial code | Function code | Music folder number | Standby | Loudness | Isomorphic value | End code |
| 0x01         | 0x51          | Data1               | 0x00    | Data2    | XOR              | 0x02     |

### III. Description:

#### 3.1. Data1 Music folder number (1 byte, range 0x00~0xFF)

| Data1 value | Corresponding music folder number |
|-------------|-----------------------------------|
| 0x00        | stop playing                      |
| 0x01        | AW001                             |
| 0x02        | AW002                             |
| 0x0F        | AW015                             |
| 0x1F        | AW031                             |
| ...         | ...                               |
| 0xFF        | AW255                             |

#### 3.2. Data2 Volume (1 byte, range 0x00~0x1c, total 29 levels)

| Data2 value | Corresponding volume level |
|-------------|----------------------------|
| 0x00        | 0                          |
| 0x01        | 1                          |
| 0x02        | 2                          |
| 0x0F        | 15                         |
| ...         | ...                        |
| 0x1C        | 28                         |

#### 3.3. XOR Isotropic or Alternative Values

The value obtained from the differentiation of the previous five bytes.

### IV. Music alarm reply PLC data format:

| 0            | 1             | 2                   | 3       | 4        | 5                | 6        |
|--------------|---------------|---------------------|---------|----------|------------------|----------|
| Initial code | Function code | Music folder number | Standby | Loudness | Isomorphic value | End code |
| 0x01         | 0x52          | Data1               | 0x00    | Data2    | XOR              | 0x02     |

Note: The PLC must receive reply data from the music alarm every time it sends a main command.

Example: PLC send 0x01 0x51 0x02 0x00 0x1c 0x4e 0x02

i.e. play the music in folder AW002 at volume 28.

Alarm reply: 0x01 0x52 0x02 0x00 0x1c 0x4d 0x02

**Note: The default playback mode is loop in folder.**

## Modbus protocol (Customer request only)

(RS232/RS485)

- 1、Serial port parameters: Baud rate: 9600 Data bits: 8 Stop bit: 1 Parity bit: none
- 2、Communication protocol: Modbus RTU slave
- 3、Slave address: 0x01 by default, if you need to modify, please specify in advance (users can not modify)
- 4、Function code: temporarily only support function code **03 06 10**
5. Modbus register address table:

| Register address | Number of registers | Byte count | Data type                                        | Functional Description                   |
|------------------|---------------------|------------|--------------------------------------------------|------------------------------------------|
| 0001             | 1                   | 2          | 16-bit positive integer                          | None                                     |
| 0002             | 1                   | 2          | 16-bit positive integer                          | None                                     |
| 0003             | 1                   | 2          | 16-bit positive integer (low bits 0 to 255 only) | Play the music in the folder AW001~AW255 |
| 004              | 1                   | 2          | 16-bit positive integers (0-28)                  | Volume level 0~28                        |
| .....            | .....               | .....      | .....                                            | .....                                    |

6. Description of the music file: **Register [3]** Corresponding music folder number (1 byte, range 0x00~0xFF)

| <b>Register [3]</b> numerical value | Corresponding music folder number |
|-------------------------------------|-----------------------------------|
| 0x0000                              | stop playing                      |
| 0x0001                              | AW001                             |
| 0x0002                              | AW002                             |
| 0x000F                              | AW015                             |
| 0x001F                              | AW031                             |
| ...                                 | ...                               |
| 0x00FF                              | AW255                             |

7. Volume Description:**Register [4]**

| <b>Register [4]</b> numerical value | Corresponding volume level |
|-------------------------------------|----------------------------|
| 0x0000                              | 0                          |
| 0x0001                              | 1                          |
| 0x0002                              | 2                          |
| 0x000F                              | 15                         |
| ...                                 | ...                        |
| 0x001C                              | 28                         |

8. Example: Play folder AW004, register [3] write 4, Modbus function code 06

PLC sends: 01 06 00 03 00 04 78 09

Alarm return: 01 06 00 03 00 04 78 09

| Action  | Slave address | Function Code | Start Address High | Start Address Low | Register Address High | Register Address Low | CRC | CRC |
|---------|---------------|---------------|--------------------|-------------------|-----------------------|----------------------|-----|-----|
| Sending | 01            | 06            | 00                 | 03                | 00                    | 04                   | 78  | 09  |
| Action  | Slave Address | Function Code | Start Address High | Start Address Low | Register Address High | Register Address Low | CRC | CRC |
| Return  | 01            | 06            | 00                 | 03                | 00                    | 04                   | 78  | 09  |

**Note: The default playback mode is Loop in Folder.**